

FCS Assignment

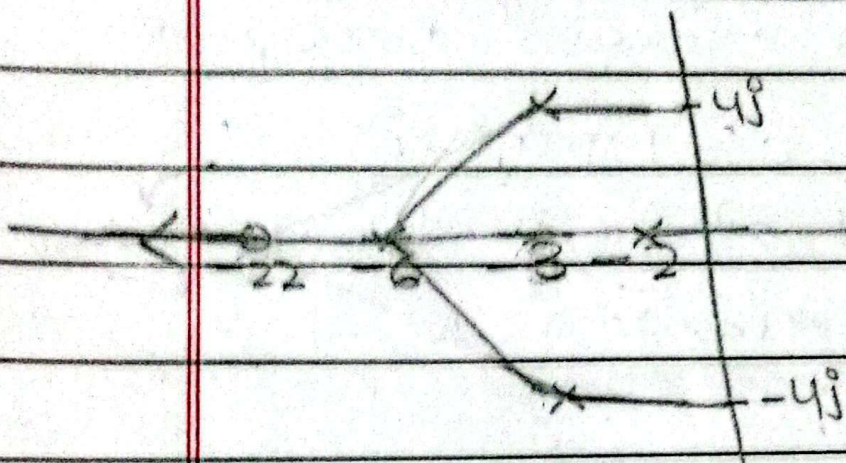
22L-6234

$$G(s) = \frac{s+22}{(s+6)(s+2)(s+3+4j)(s+3-4j)}$$

zeros = -22

Poles = -6, -2, -3-4j, -3+4j

I: Draw on real axis



II: Asymptotes

$$\sigma_a = \frac{\pm 180(2k+1)}{n-m} = \frac{\pm 180(2k+1)}{4-1}$$

$$= \frac{\pm 180(2k+1)}{3} \quad \text{for } k=0,1,2$$

$$= 60^\circ, 180^\circ, 300^\circ$$

$$\zeta_a = \frac{\sum \text{pole} - \sum \text{zero}}{n - m}$$

$$= \frac{(-6) + (-2) + (-3 + 4j) + (-3 - 4j) + (-22)}{3}$$

$$= -12$$

III : Breakpoints

$$1 + k(s+22) = 0$$

$$(s+6)(s+2)(s+3+4j)(s+3-4j)$$

$$k = - \frac{(s+6)(s+2)(s+3+4j)(s+3-4j)}{s+22} \quad \text{--- (1)}$$

$$k = - \frac{(s^2+8s+12)(s^2+16)}{s+22}$$

$$k = - \frac{(s^2+8s+12)(7s)}{s+22}$$

$$k = - \frac{(s^4+14s^3+88s^2+272s+300)}{s+22}$$

$$\frac{dk}{ds} = - \frac{(s+22)(4s^3+4s^2+170s+272) - (s^4+14s^3+88s^2+272s+300)}{(s+22)^2}$$

$$\frac{dk}{ds} = - \frac{[4s^4 + 4s^3 + 170s^2 + 272s + 88s^3 + 88s^2 + 3740s + 5984 - s^4 - 14s^3 - 85s^2 + 272s + 300]}{(s+22)}$$

$$= - \frac{[3s^4 + 78s^3 + 173s^2 + 3740s + 5684]}{(s+22)}$$

$$\frac{dk}{ds} = 0$$

$$3s^4 + 78s^3 + 173s^2 + 3740s + 5684 = 0$$

$$s_1 = -25.54 \text{ — on R.L}$$

$$s_2 = 0.55 + 6.88j$$

$$s_3 = 0.55 - 6.88j$$

$$s_4 = -1.56$$

} Not on R.L

→ put in ① for K

$$K = - \frac{[(-25.54+6)(-25.54+2)(-25.54+3+4j)(-25.54+3-4j)]}{(-25.54+22)}$$

$$K = 68005.033 + 15.59j$$

IV: Determine jw crossing

$$\frac{1 + k(s+22)}{(s+2)(s+3)(s+3+4j)(s+3-4j)} = 0$$

$$1 + \frac{k(s+22)}{s^4 + 14s^3 + 85s^2 + 272s + 300} = 0$$

$$s^4 + 14s^3 + 85s^2 + 272s + 300 + sk + 22k = 0$$

$$s^4 + 14s^3 + 85s^2 + (272+k)s + 300 + 22k = 0$$

s^4	1	85	$300 + 22k$
s^3	14	$272 + k$	0
s^2	$\frac{-918 + k}{14}$	$300 + 22k$	0
s^1	$\frac{(-918 + k)(272 + k)}{14}$ $\frac{-14(300 + 22k)}{14}$ $\frac{-918 + k}{14}$	0	0
s^0	$300 + 22k$	0	0

$$300 + 22k = 0$$

$$k = \frac{-300}{22} = -13.63 \text{ (-ve so no roots on } \omega)$$

V: Angle of arrival and angle of departure

For $s = -3 \pm 4j$ angle of departure

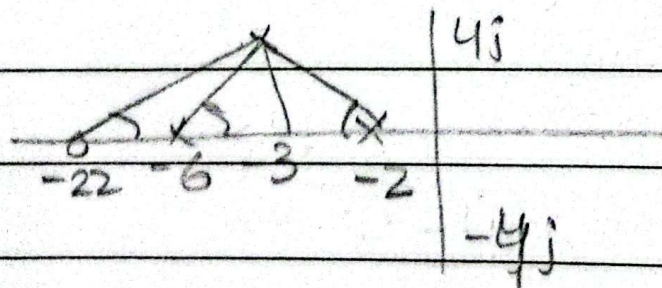
1- Pole at $s = -6$: $\theta = \tan^{-1}\left(\frac{4}{3}\right) = 53.13^\circ$

2- Pole at $s = -2$: $\theta = \tan^{-1}\left(\frac{4}{1}\right) = 75.96^\circ$

3- Zero at $s = -22$: $\theta = \tan^{-1}\left(\frac{4}{19}\right) = 11.89^\circ$

$$\sum_{\text{pole}} = 53.13^\circ + 75.96^\circ = 129.09$$

$$\text{Angle of departure} = 180^\circ - (129.09 - 11.89) = 62.8^\circ$$



Part 2: PID controller

Given: overshoot = 12% = 0.12

$$T_s = 8 \text{ sec}$$

No steady state error

1- Damping ratio:

$$0.12 = e^{\left(\frac{-\pi \zeta}{\sqrt{1-\zeta^2}}\right)} \times 100\%$$

$$0.12 = e^{\left(\frac{-\pi \zeta}{\sqrt{1-\zeta^2}}\right)}$$

$$\frac{-\pi \zeta}{\sqrt{1-\zeta^2}} = \ln(0.12) \Rightarrow \boxed{\zeta = 0.555}$$

2- Natural frequency:

$$T_s = \frac{4}{\zeta \omega_n} = 8$$

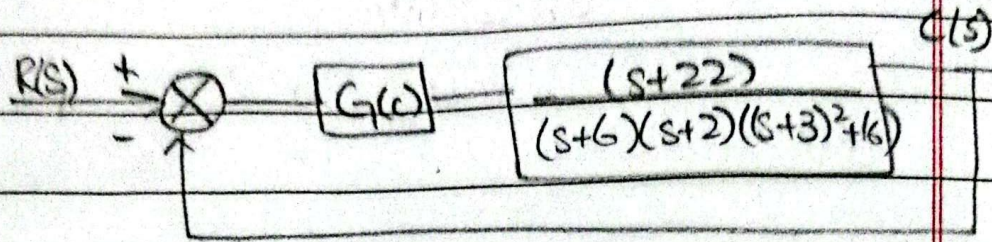
$$\frac{8}{4} = \frac{1}{\zeta \omega_n} \Rightarrow \boxed{\omega_n = 0.897}$$

3- Pole location:

$$s = -\zeta \omega_n \pm j \omega_n \sqrt{1-\zeta^2}$$

$$\boxed{s = -0.5 \pm j0.72}$$

PID controller



$$G(s) = G(s) \cdot \frac{(s+22)}{(s+6)(s+2)((s+3)^2+16)}$$

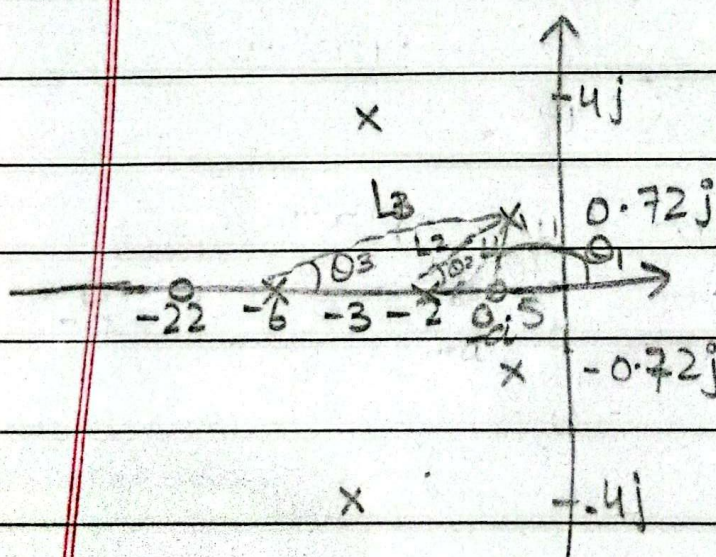
where $G(s) = \frac{k(s+a)(s+b)}{s}$

$$= \frac{k(s+22)(s+a)(s+b)}{s[(s+6)(s+2)((s+3)^2+16)]}$$

→ Type 1 system

Now,

desired pole at $-0.5 \pm j0.72$



$$G_{PD} = k(s+a)$$

zero at -a

(yeh meri end pr neat bha dngi)

$$\sum \theta_2 - \sum \theta_p = -180^\circ$$

$$\theta_1 - (\theta_2 + \theta_3) = -180^\circ$$

$$\theta_2 = \tan^{-1} \left(\frac{0.72}{2 - 0.5} \right) = 25.64^\circ$$

$$\theta_3 = \tan^{-1} \left(\frac{0.72}{6 - 0.5} \right) = 7.45^\circ$$

$$\theta_1 = 180^\circ - \tan^{-1} \left(\frac{0.72}{0.5 - a} \right)$$

$$\left(180 - \tan^{-1} \left(\frac{0.72}{0.5 - a} \right) \right) - (25.64^\circ + 7.45^\circ) = -180^\circ$$

$$180^\circ - \tan^{-1} \left(\frac{0.72}{0.5 - a} \right) - 33.09 = -180^\circ$$

$$-\tan^{-1} \left(\frac{0.72}{0.5 - a} \right) = -180^\circ - 146.91$$

$$\tan^{-1} \left(\frac{0.72}{0.5 - a} \right) = 326.91$$

$$\frac{0.72}{0.5 - a} = 0.65$$

$$\frac{0.72}{0.65} - 0.5 = -a$$

$$a = -0.60$$

For gain:

$$K_{\text{overall}} = \frac{\pi L_P}{\pi L_Z} = \frac{L_2 L_3}{L_1}$$

$$L_2 = \sqrt{(0.72)^2 + (1.5)^2} = 1.66$$

$$L_3 = \sqrt{(0.72)^2 + (5.5)^2} = 5.54$$

$$L_1 = \sqrt{(0.72)^2 + (0.1)^2} = 0.72$$

$$K_{\text{overall}} = 12.77$$

$$G_{PD} = 12.77(s + 0.60)$$

$$G_L = \frac{K(s+a)(s+b)}{s}$$

$s \longrightarrow$ pole at 0

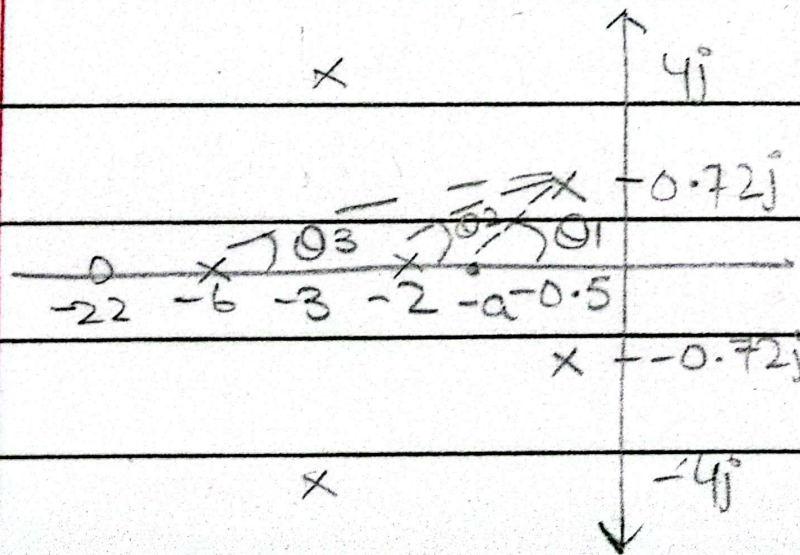
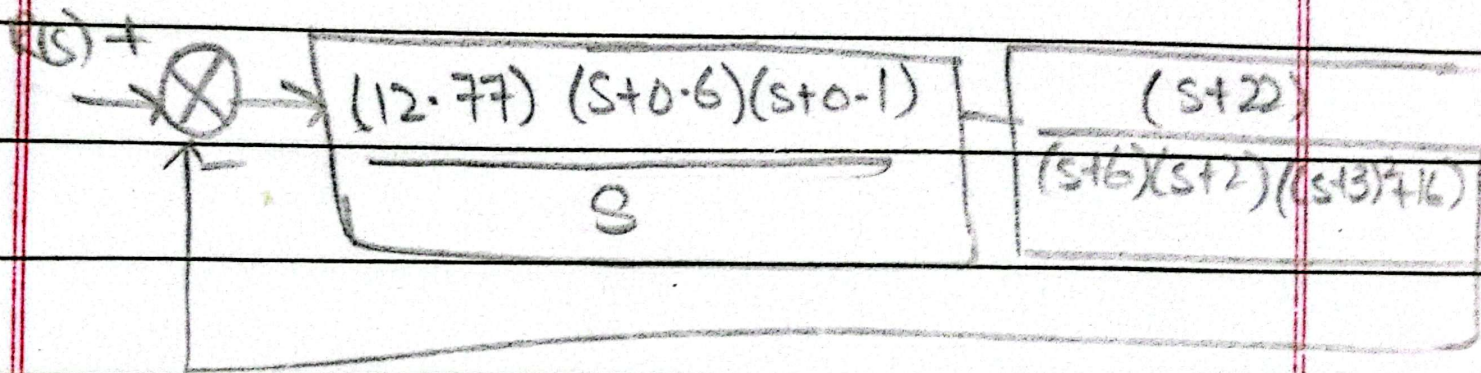
$$\text{Let } b = -0.1$$

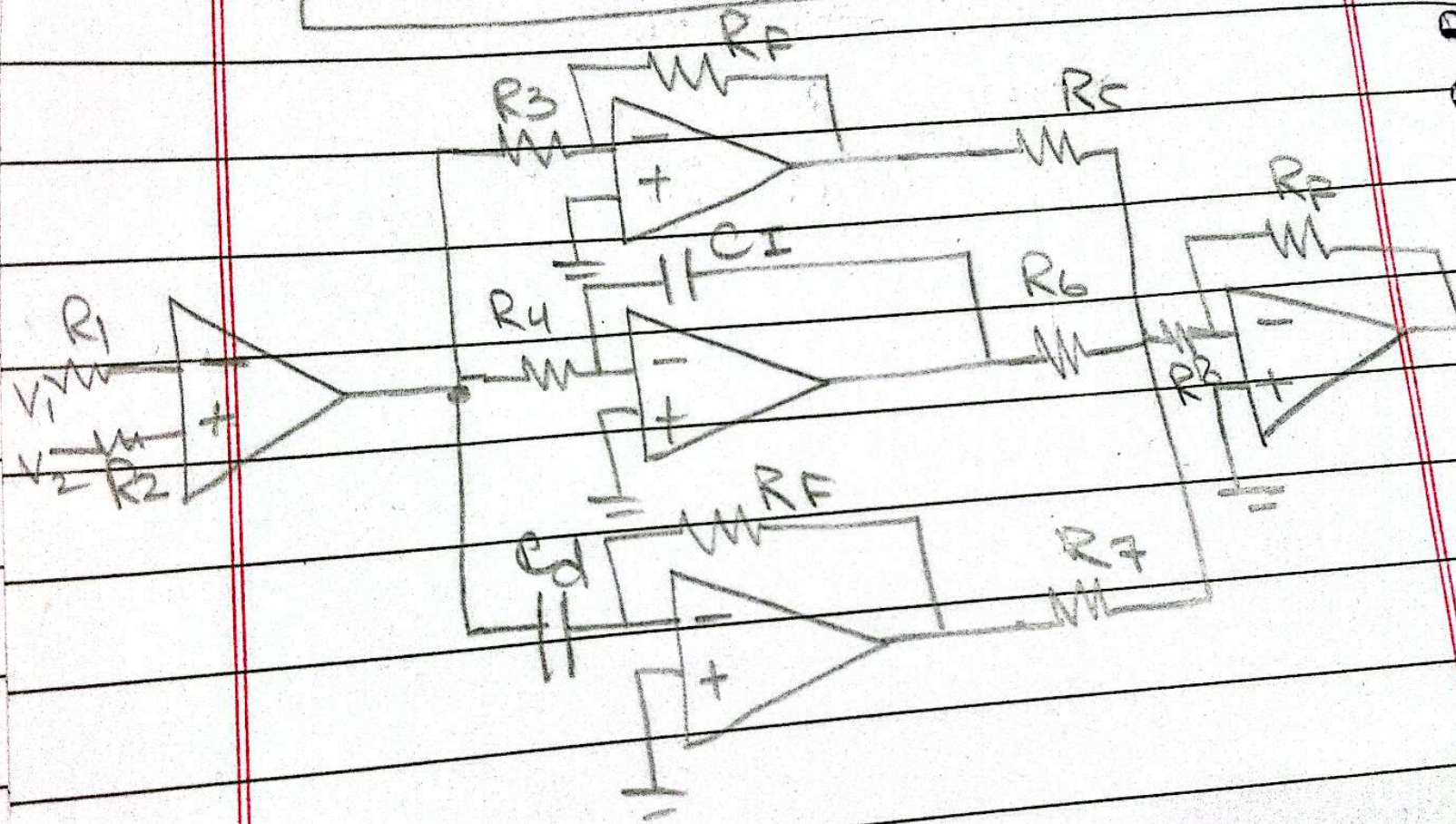
$$G(c) = \frac{12.77(s + 0.6)(s + 0.1)}{s}$$

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Part C





Setting time
Desired po
 $G(s)$

Lead-Lag Controller

$$T_s = 4 \text{ sec}$$

$$\%OS = 12\%$$

$$\zeta = 0.55$$

$$SSE = 15\%$$

$$\omega_n = 0.897$$

$$\text{Desired poles} = -0.5 \pm j0.72$$

Design lead:

$$G_{open} = \frac{G_c (s+22)}{(s+6)(s+2)((s+3)^2+16)}$$

$$G_c = \frac{k(s+z_{lead})}{(s+p_{lead})}$$

Improving steady state error by 15%.

$$\frac{1}{0.85} \approx 1.176$$

Lag compensator:

$$C_{lag}(s) = \frac{s+z_{lag}}{s+p_{lag}}, \quad \frac{z_{lag}}{p_{lag}} = \frac{1}{1.176} \approx 0.85$$

$$z_{lag} = 0.05 \Rightarrow p_{lag} = 0.0588$$

Back to Lead compensator:

$$s = -0.5 \pm j0.72$$

Assume

Lead zero at $z_{lead} = 1$

Trial pole at $p_{lead} = 4.5$

$$C_{lead}(s) = \frac{s+1}{s+4.5}$$

So final compensator

$$C(s) = \frac{(s+22)}{(s+6)(s+2)((s+3)^2+16)} \cdot \frac{s+1}{(s+4.5)} \cdot \frac{s+0.05}{(s+0.0588)}$$

Part 5 A/SC1

